

10.18

(a)

The percentage of mothers with some college education is 12.15%.

The percentage of fathers with some college education is 11.68%.

(b)

(1)

The correlation coefficient between *EDUC* and *MOTHERCOLL* is 0.36, while the correlation coefficient between *EDUC* and *FATHERCOLL* is 0.40. These correlation coefficients indicate a positive relationship between *EDUC* and parents' college education, but it is not very strong.

(2)

According to the conditions for instrumental variables :

1. The instrumental variable should not have a direct effect on the dependent variable (y_i) and therefore should not be included as an explanatory variable in the model.
2. It should not be correlated with the error term of the regression (i.e., it should be exogenous).
3. It should be strongly correlated (or at least not weakly correlated) with the endogenous variable (x_i).

Given the context provided, *MOTHERCOLL* and *FATHERCOLL* are more likely to satisfy condition two. This is because they are binary variables indicating whether a parent has some college education, which are less susceptible to endogeneity concerns compared to continuous variables like *MOTHEREDUC* and *FATHEREDUC*. These binary variables are less likely to be influenced by contemporaneous factors affecting the dependent variable. Therefore, we can conclude that *MOTHERCOLL* and *FATHERCOLL* are better instrumental variables than the *MOTHEREDUC* and *FATHEREDUC*.

(c)

The wage equation in Example 10.5 is $\ln(\text{WAGE}) = \beta_1 + \beta_2 \text{EXPER} + \beta_3 \text{EXPER}^2 + \beta_4 \text{EDUC} + e$.

First, we take *MOTHERCOLL* to substitute *EDUC*, and get first-stage equation is $\text{EDUC} = \gamma_1 + \theta_1 \text{MOTHERCOLL} + v$.

Next, we take first-stage equation into original model to get second-stage equation is $\ln(\text{WAGE}) = \beta_1 + \beta_2 \text{EXPER} + \beta_3 \text{EXPER}^2 + \beta_4 \text{MOTHERCOLL} + e^*$.

The 95% interval estimate of β_4 is $[-0.0012, 0.1533]$.

(d)

The first-stage equation is $X = \gamma_1 + \theta_1 Z + v$

which is $EDUC = \gamma_1 + \theta_1 MOTHERCOLL + v$

Estimate the first-stage equation by OLS and obtain the fitted value: $\hat{X} = \hat{\gamma}_1 + \hat{\theta}_1 Z$ which is $\widehat{EDUC} = \hat{\gamma}_1 + \hat{\theta}_1 MOTHERCOLL$.

The F-test statistic for the hypothesis that MOTHERCOLL has no effect on EDUC is 63.21602.

$\because 63.21602$ is greater than the rule of thumb value of 10. $\therefore$ We reject the null hypothesis that the instrumental variable MOTHERCOLL is weak.

(e)

The 95% interval estimate for the coefficient of EDUC using MOTHERCOLL and FATHERCOLL as the instrumental variables is $[0.02752, 0.1482]$. It is narrower than the one in part (c).

(f)

The first-stage equation is $X = \gamma_1 + \theta_1 Z_1 + \theta_2 Z_2 + v$

which is $EDUC = \gamma_1 + \theta_1 MOTHERCOLL + \theta_2 FATHERCOLL + v$.

Therefore, we get that $\hat{X} = \hat{\gamma}_1 + \hat{\theta}_1 Z_1 + \hat{\theta}_2 Z_2$

which is $\widehat{EDUC} = \hat{\gamma}_1 + \hat{\theta}_1 MOTHERCOLL + \hat{\theta}_2 FATHERCOLL$.

The F-test statistic of the joint significance of the two instrumental variables is 57.59666.

$\because 57.59666$ is greater than the rule of thumb value of 10.

$\therefore$ We reject the null hypothesis that the instrumental variables MOTHERCOLL and FATHERCOLL are weak.

(g)

When we have surplus instrument, we can use sargan test to estimate that the instrument is valid or not.

Therefore, we can have a null hypothesis that the surplus instrument is valid, i.e. $\text{cov}(z_i, e_i) = 0$, from using NR^2 , if NR^2 is lower than critical value, then we know that the null hypothesis can not be rejected.

In this case, the NR^2 is 0.2375851. This test statistic has a $\chi^2(1)$ distribution under the null hypothesis that the surplus instrument is valid, and the 5% critical value is 3.841.

\(\therefore\) We can not reject the null hypothesis. It means that the instrumental variables z_i is exogeneity from the residual e_i .

10.20

(a)

Microsoft's estimated beta is 1.2018, with a standard error of 0.122152. The t-value is 9.839, and the p-value is very small ($< 2e-16$). This means that the relationship is significant. Therefore, the Microsoft stock is relatively risky as its beta is greater than one.

(b)

The variable *RANK* has no obvious causal effect on Microsoft's return, so it satisfies condition IV1. It is probable to believe that the regression error in time t is not correlated to the ranked values, so the classification is probably exogenous to satisfy IV2. When $(r_m - r_f)$ is larger, the ranking should be higher. Hence we can conclude that IV3 is satisfied.

The $R^2 = 0.9126$ is very large. The t-value is 43.10 which corresponds to $F = 1858$, which is huge. Therefore, *RANK* seems to be an extremely strong IV, if it is valid.

(c)

Coefficients	Estimate	Std. Error	t value	P-value
$\hat{v}$	-0.874599	0.428626	-2.040	0.0428

This is the Hausman test for endogeneity. The t-statistic on $\hat{v}$ is -2.04 with a p-value = 0.043. It is not significant at the 1% level. Therefore, we cannot reject the null hypothesis that the market return is exogenous at the 1% level.

(d)

The estimate results are as followed :

Coefficients	Estimate	Std. Error	t value	P-value
RP	1.2783182	0.1317427	9.7031	<2e-16

And the 95% interval estimate of RP is :

Coefficients	2.5%	97.5%
RP	1.027421458	1.52921503

If there is a measurement error problem (and the error is random), the OLS estimator suffers attenuation bias and is biased downward. The coefficient of RP in IV regression (1.2783) is slightly larger than the original OLS one (1.2018), which is what we would expect. The 95% interval estimate is now [1.0274, 1.5292]. We would still reject the null hypothesis that Microsoft's beta equals to 1.

(e)

The F-test statistic for the joint significance of these variables is 951.3. Based on the F-test we would conclude the IV are not weak. The R^2 is 0.9149. *RANK* remains strongly significant but *POS* is not significant at the 1% level. If we are willing to accept a 5% test then we can conclude the IV are not weak.

(f)

The t-value on the first stage residuals is -2.21 with a p-value of 0.029. It is not significant at the 1% level. Therefore, we cannot reject the null hypothesis that the market return is exogenous at the 1% level of significance.

(g)

Coefficients	Estimate	Std. Error	t value	P-value
MKTRET	1.283118	0.127866	10.035	<2e-16

(1) Estimates an IV regression using the *ivreg* function the dependent variable (*msft-riskfree*) and the endogenous variable (*MKTRET*). *RANK* and *POS* are instrumental variables.

(2) The IV coefficient estimate 1.28 is larger than the OLS estimate from part (a) 1.20. The IV estimate is slightly larger, which is what we would expect.

(h)

1. Estimates an IV regression using the *ivreg* function the dependent variable (*msft-riskfree*) and the endogenous variable (*MKTRET*). *RANK* and *POS* are instrumental variables.

2. Runs an (OLS) regression using the extracted residuals as the dependent variable and the (*RANK* and *POS*) as independent variables.

3. Multiplies the R-squared value by the number of observations.

The test statistic $NR^2 = 0.56$

The rejection region is $\{X^2: X^2 > 3.84\}$

Therefore, we cannot reject the $H_0: cov(RANK, e) = 0, cov(POS, e) = 0$
It means that the surplus IVs ($RANK$ and POS) are valid instruments.